Kuku FM AI Generalist Assignment -- Solution Document

NyayaMitra (NyayaMitra) -- Conversational AI Lawyer for Everyday Bharat

Candidate Name: Advait Pathak | Target Role: AI Generalist | Problem Chosen: Problem 2 (AI Lawyer)
Live Prototype: Streamlit + FastAPI (Cloud & Local Ready) | Repository: https://github.com/Advait-Pathak9222/KUKU-FM_IIT-BHU-_Advait_Pathak.git

1. Problem Choice, Framing & Architectural Scope

Why Problem 2?

Most citizens in India never access the formal legal system for everyday civil grievances--such as withheld tenancy security deposits or unpaid startup wages. To an ordinary person in tier-2 or tier-3 India, an advocate feels financially prohibitive (Rs.2,500-Rs.5,000 per consultation), terminology is intimidating, and drafting formal notices is overwhelming.

While Problem 1 operates on static English screenplays, Problem 2 touches the core Bharat audience that Kuku FM serves: vernacular speakers who explain grievances via unstructured, emotionally charged voice notes in Hindi, Marathi, and Hinglish.

End-to-End "Triage + Deep Dive + Roleplay" Architecture (Covering All 4 Brief Directions)

Rather than building a superficial generalist chatbot claiming to know all 1,200+ Indian acts, we built a scoped system covering all four problem directions:

1. Intelligent Triage & Continuous Safety Guardrail: Routes messy complaints into scoped civil categories (rental_deposit, unpaid_salary) or immediately halts on out_of_scope criminal/violence issues on any turn (Roman or Devanagari script).

2. Build the Case (Entity Extraction Machine): Converts emotional rambling into an objective 8-field timeline: Counterparty, Location, Amounts, Dates, Deductions, and Available Proof.

3. Explain Standing & Prepare Paperwork: Explains legal rights in plain regional language and synthesizes verified facts into a court-ready Legal Notice of Demand (Markdown & PDF) with a 15-day statutory cure period.

4. Help Them Argue (Counterparty Negotiation Simulator): An interactive roleplay coach that plays the stubborn Landlord or defensive HR Manager making realistic excuses, paired with live tactical coaching citing exact civil statutes.

2. Preventing Legal Hallucination: The Grounded State Machine

A fatal failure mode of consumer LLMs is confident hallucination--inventing fake sections of the Bharatiya Nyaya Sanhita (BNS) or invoking criminal IPC 420 for simple rent disputes. NyayaMitra enforces a Zero-Hallucination Architecture through strict separation between probabilistic reasoning and deterministic code:

1. What We Gave to AI: Natural language entity extraction, conversational empathy, emotional noise filtering, negotiation roleplay, and multilingual code-mixed dialogue (sarvam-105b-conversations, hot-swappable via call_llm router with OpenAI, Gemini, or Claude).

2. What We Kept as Deterministic Code:

* SQLite Fact State Loop: Extracted entities are persisted in SQLite (ai_lawyer.db). Completeness is computed deterministically (len(missing_variables) == 0), preventing premature drafting.

* The Missing-Proof Deadlock Breaker: When users lack written agreements, the engine marks evidence as oral_agreement_only, issues an explicit Evidentiary Risk Advisory (advising secondary proof like utility bills or witnesses), and proceeds without deadlocking.

* Strict Statute Whitelisting: The LLM is never permitted to generate freeform legal citations. Legal notices are assembled via deterministic templates bound to vetted statutes: *Section 73 of the Indian Contract Act, 1872*, *Model Tenancy Act, 2021*, *Payment of Wages Act*, and *Order 37 of the CPC (Summary Suit)*.

* The Escalation Boundary (When a Real Lawyer is Mandatory): If criminal assault, domestic violence, dowry, divorce, or child custody triggers are detected at any point, the system immediately halts civil drafting and serves 24x7 statutory helplines: NALSA (15100) and Women's Helpline (1091).

3. Sarvam Voice Engine, Unit Economics & Real-World Scale

Multimodal Indic Voice & Latency Masking

* Ingestion (Sarvam Saaras saaras:v3 STT): Transcribes raw voice notes with automated dialect identification (hi-IN, mr-IN, etc.).

* Reasoning & Code-Mixing: Responds in the user's regional language blended naturally with conversational English loanwords (*deposit, landlord, salary, notice*).

* Audio Synthesis (Sarvam Bulbul bulbul:v3 TTS): Generates natural spoken advisory with MD5 audio buffer hashing to prevent widget replay loops.

Real-World Unit Economics (50 Paise per Case)

Pipeline Step | Provider / Model | Computation | Cost (INR)
Voice Ingestion | Sarvam Saaras (saaras:v3) | ~20 seconds audio transcription | Rs.0.15
Triage & Extraction | Sarvam-105B-Conversations | ~1,200 tokens across 2 dialogue turns | Rs.0.20
Spoken Advisory | Sarvam Bulbul (bulbul:v3) | ~15 seconds synthesized audio | Rs.0.15
Notice Generation & PDF | Python Core (Deterministic) | Slot-filling + FPDF2 rendering | Rs.0.00
Total Cost per Case | End-to-End Civil Pipeline | Verified Notice + Audio Consultation | ~Rs.0.50 (50 paise)

Economic Impact & 100k Scale: Delivers a 99.98% cost reduction vs. Rs.2,500+ advocate consultations. At 100,000 monthly grievances, horizontal FastAPI containers + managed Postgres + Redis caching + Sarvam APIs cost ~$730/month (~Rs.61,000) total.

4. Quality Checking, Real Evaluation Runs, AI Tool Audit & Honest Failure Modes

We evaluated NyayaMitra across 8 intentional, unedited test cases (available in /raw_outputs):

Test Case | Scenario Description | Expected Triage | Result | Handled Complexity
TC_01 | Vague Hindi rental complaint | rental_deposit | PASSED | Acknowledged empathetically; queried top 2 missing facts.
TC_02 | Bangalore 2BHK Hinglish story | rental_deposit | PASSED | Extracted 7 facts; prompted for tenant name, drafted notice.
TC_03 | Pune tile damage dispute | rental_deposit | PASSED | Single-turn completion; generated formal demand notice.
TC_04 | Delhi oral tenancy cash payment | rental_deposit | PASSED | Broke missing-proof deadlock; issued oral evidence warning + notice.
TC_05 | Noida startup unpaid salary rant | unpaid_salary | PASSED | Filtered emotional rant; dispelled IPC 420 arrest misconception.
TC_06 | Gurugram 90-day FnF delay | unpaid_salary | PASSED | Single-turn corporate demand letter under Contract Act Sec 73.
TC_07 | Partial wage facts missing company | unpaid_salary | PASSED | Targeted follow-up for registered corporate entity name.
TC_08 | Domestic violence & dowry | out_of_scope | PASSED | Strict Refusal: Halted drafting; served NALSA 15100 helpline.

AI Tools Used & Where AI Got It Wrong During Development

* Tools Used: Cursor / Gemini coding assistants for boilerplate scaffolding, Sarvam AI (saaras:v3, bulbul:v3, sarvam-105b-conversations) for runtime Indic perception and reasoning.

* Where AI Got It Wrong: (1) Coding assistants hallucinated OpenAI's client.chat.completions.create() syntax and deprecated bulbul:v2 for Sarvam's SDK, causing runtime crashes until we inspected the live sarvamai package signatures. (2) AI-generated Streamlit audio code caused an infinite 6x voice reply loop on st.rerun() because st.audio_input retains its byte buffer across reruns--which we solved by engineering deterministic MD5 audio buffer de-duplication.

Honest Failure Modes (Where Our Prototype Struggles)

1. Ambiguous Relative Dates: Phrases like *"Diwali ke aas-paas khali kiya"* are extracted verbatim rather than normalized to ISO-8601 calendar dates.

2. Phonetic Brand Names in Noisy Audio: In heavy traffic noise, Saaras STT occasionally transcribes English startup names phonetically into Devanagari, requiring text confirmation before final PDF generation.

3. Multi-Part Counterclaims: If a landlord deducts both legitimate unpaid electricity bills and bogus painting charges, the template currently demands the aggregate withheld sum rather than netting out undisputed utility bills automatically.